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Appendix

Model with introgression

When there is introgression, some loci have a history that takes a different path through the species network. For the simplest case with one introgression event, there is one reticulation and therefore one additional "parent tree" embedded in the species network (see Hibbins and Hahn 2019 for full explication). Here we describe the expectations for introgression from species C into species B (as in Figure 2A in the main text). Other introgression events follow the same logic as this one. The additional parent tree generated by this introgression has lineages C and B sister to one another, and is defined by two split times: t_m and t_2. The time t_2 is the same as in the species tree, but now lineages B and C can coalesce starting at time t_m (Figure 2A).

This parent tree can also produce all three possible topologies. Because the BC topology is now the one that matches the parent tree, there are two histories with this topology; we denote these $BC1_2$ and $BC2_2$. The two topologies discordant with the parent tree from an introgression history are denoted AB_2 and AC_2. The expected frequencies of these topologies are:

$$E[f_{BC2_2}] = E[f_{AB_2}] = E[f_{AC_2}] = (1/3)e^{-(t_2-t_m)} \tag{1}$$

$$E[f_{BC1_2}] = 1 - e^{-(t_2-t_m)} \tag{2}$$

Our goal here is to find the expectation for D_3 (as given in equation 8 in the main text) in the presence of introgression. We therefore require the expected coalescence times t_{B-C} and t_{A-C} for each topology from the second parent tree:

$$E[t_{B-C}|BC1_2] = t_m + (1 - \frac{t_2 - t_m}{e^{(t_2-t_m)} - 1}) \tag{3}$$

$$E[t_{B-C}|BC2_2] = t_2 + 1/3 \tag{4}$$

$$E[t_{B-C}|AB_2] = E[t_{B-C}|AB_2] = t_2 + 1/3 + 1 \tag{5}$$

and

$$E[t_{A-C}|BC1_2] = t_2 + 1 \tag{6}$$

$$E[t_{A-C}|BC2_2] = E[t_{A-C}|AB_2] = t_2 + 1/3 + 1 \tag{7}$$

$$E[t_{A-C}|AC_2] = t_2 + 1/3 \tag{8}$$

The expected distances between lineages across all loci will be comprised of the average distance across trees with both introgressed and non-introgressed histories. Therefore, we must weight the contributions of each history by the admixture proportion, γ, which describes the fraction of the genome following the introgression history (with 1-γ following the species history). Combining results on the expected time to coalescence for the species history (given in the main text, and denoted with the subscript "1" here) with the expected times for the introgression history (supplementary equations 1-8), we have:

$$\begin{aligned} E[d_{B-C}] = &(1-\gamma) * 2\mu[f_{AB1_1}(t_2+1) \\ &+ f_{AB2_1}(t_2+1/3+1) + f_{BC_1}(t_2+1/3) \\ &+ f_{AC_1}(t_2+1/3+1)] + \\ &\gamma * 2\mu[f_{BC1_2}(t_m + (1 - \frac{t_2-t_m}{e^{(t_2-t_m)}-1})) \\ &+ f_{BC2_2}(t_2+1/3) + f_{AB_2}(t_2+1/3+1) \\ &+ f_{AC_2}(t_2+1/3+1)] \end{aligned} \tag{9}$$

and

$$
\begin{aligned}
E[d_{A-C}] = {} & (1-\gamma) * 2\mu[f_{AB1_1}(t_2+1) \\
& + f_{AB2_1}(t_2+1/3+1) + f_{BC_1}(t_2+1/3+1) \\
& + f_{AC_1}(t_2+1/3)] + \gamma * 2\mu[f_{BC1_2}(t_2+1) \\
& + f_{BC2_2}(t_2+1/3+1) + f_{AB_2}(t_2+1/3+1) \\
& + f_{AC_2}(t_2+1/3)] \quad (10)
\end{aligned}
$$

These expected values can be used to find the value of D_3 for any amount of introgression at any time in the past (see, for example, Figure 3 in the main text).

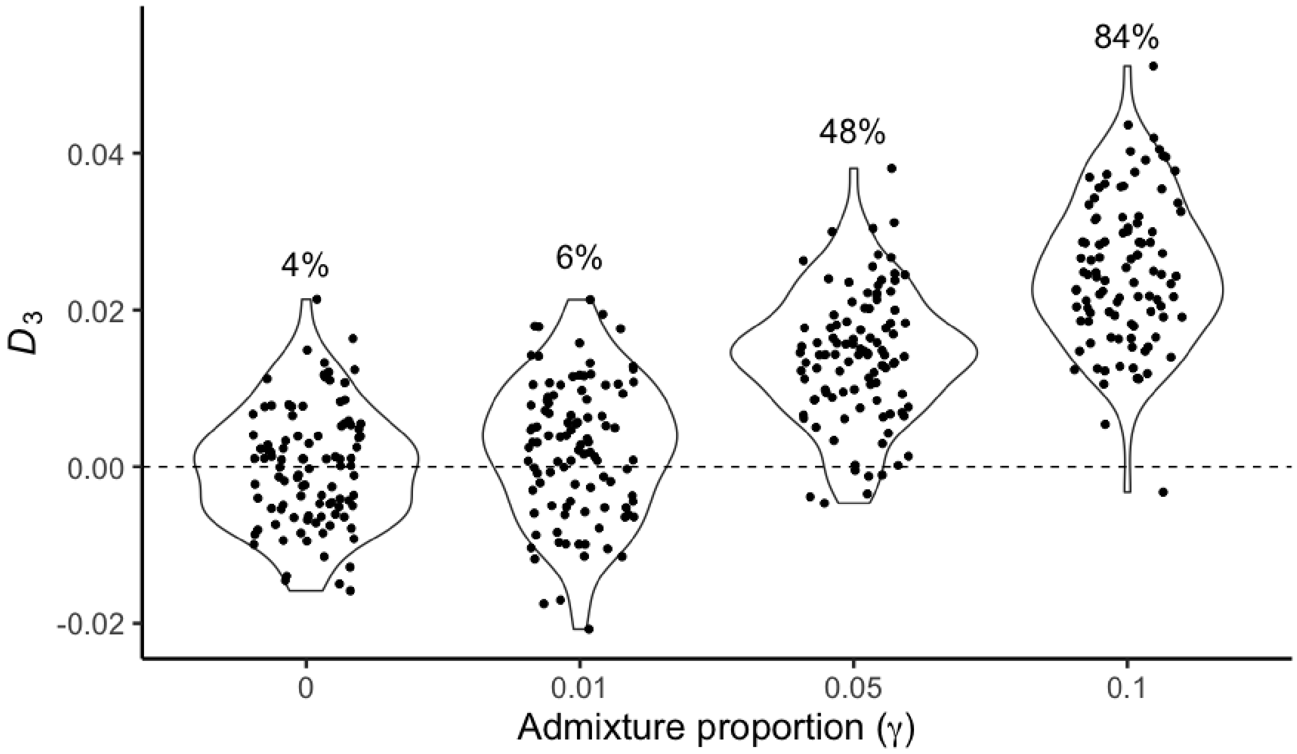

FIG. 1. Power of D_3 with the size of simulated alignments reduced to 100 kilobases (vs. 1 megabase in all others).

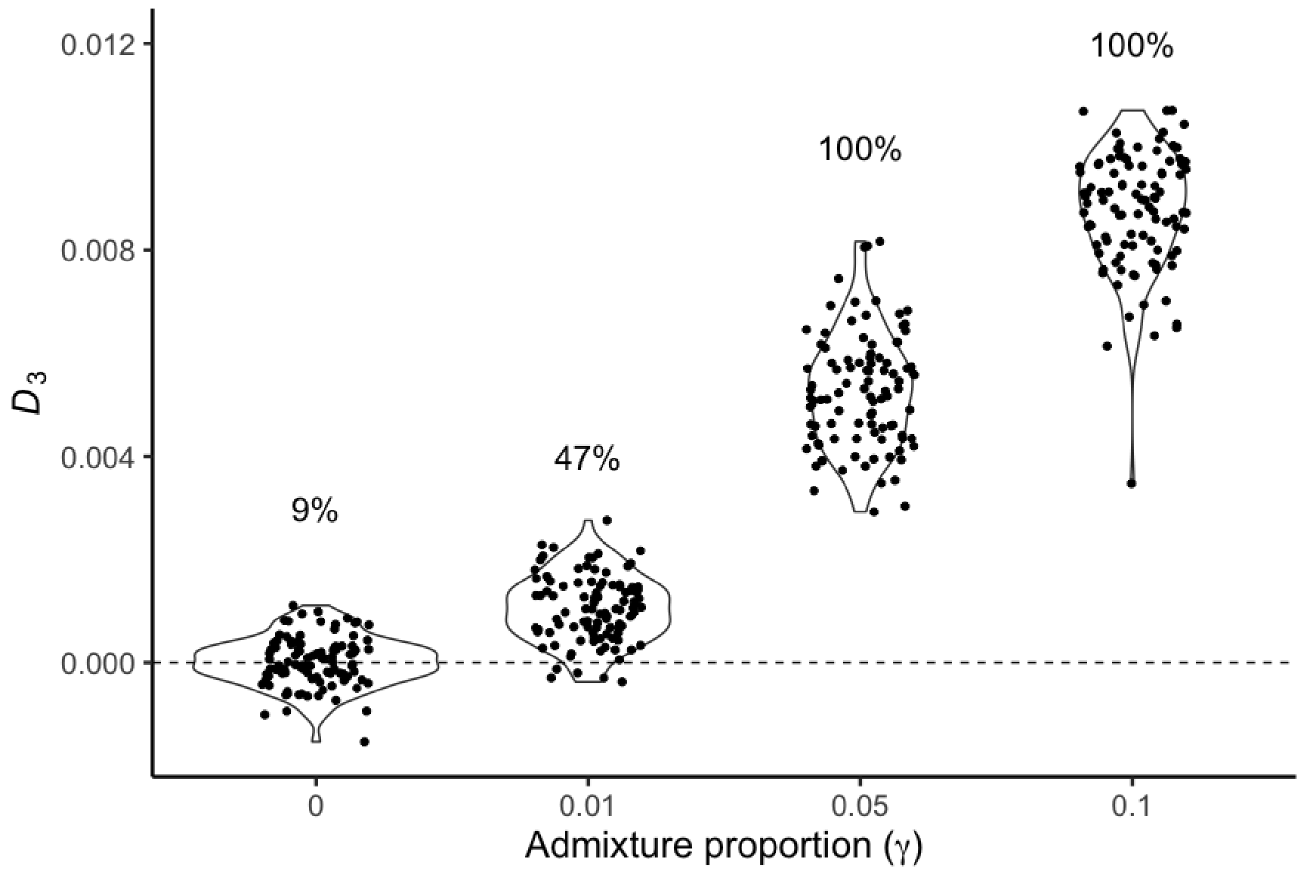

FIG. 2. Power of D_3 with θ increased to 2.

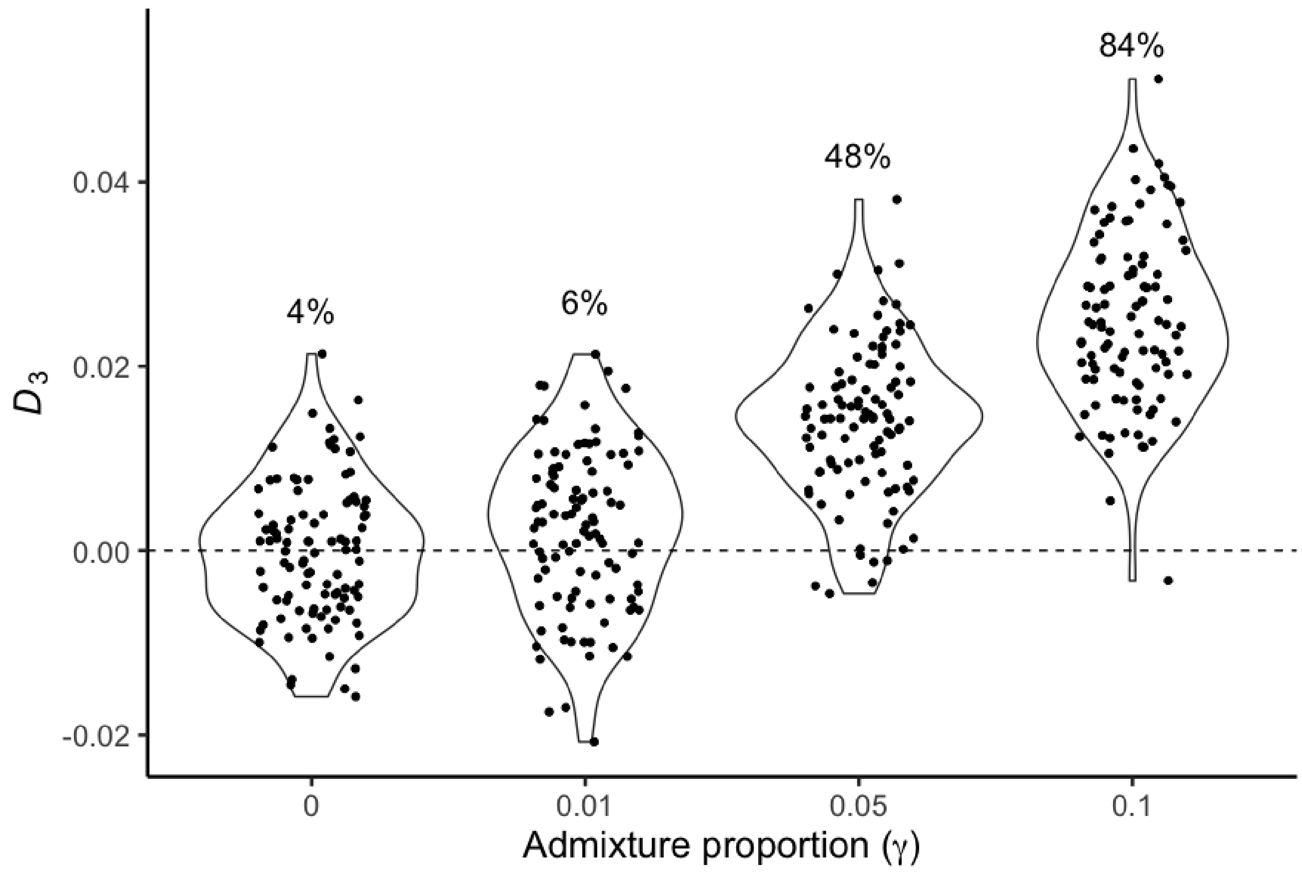

FIG. 3. Power of D_3 with interlocus variation in θ. In each simulated dataset, θ at a locus can be one of four different values. Across loci, θ averages to 0.01, the same as in all other simulations.

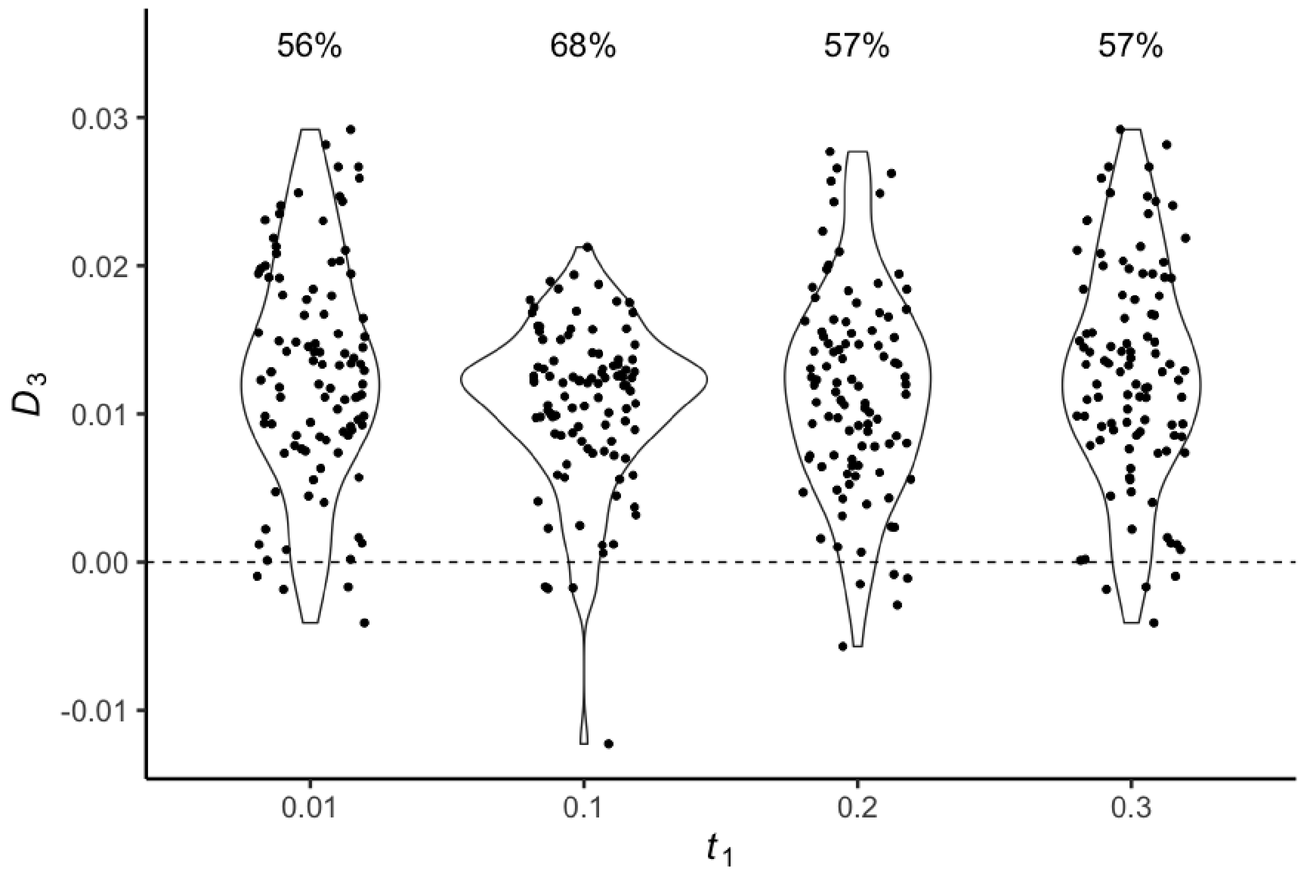

FIG. 4. Power of D_3 across four values of t_1 (in units of $4N$ generations), with $\gamma = 0.05$.

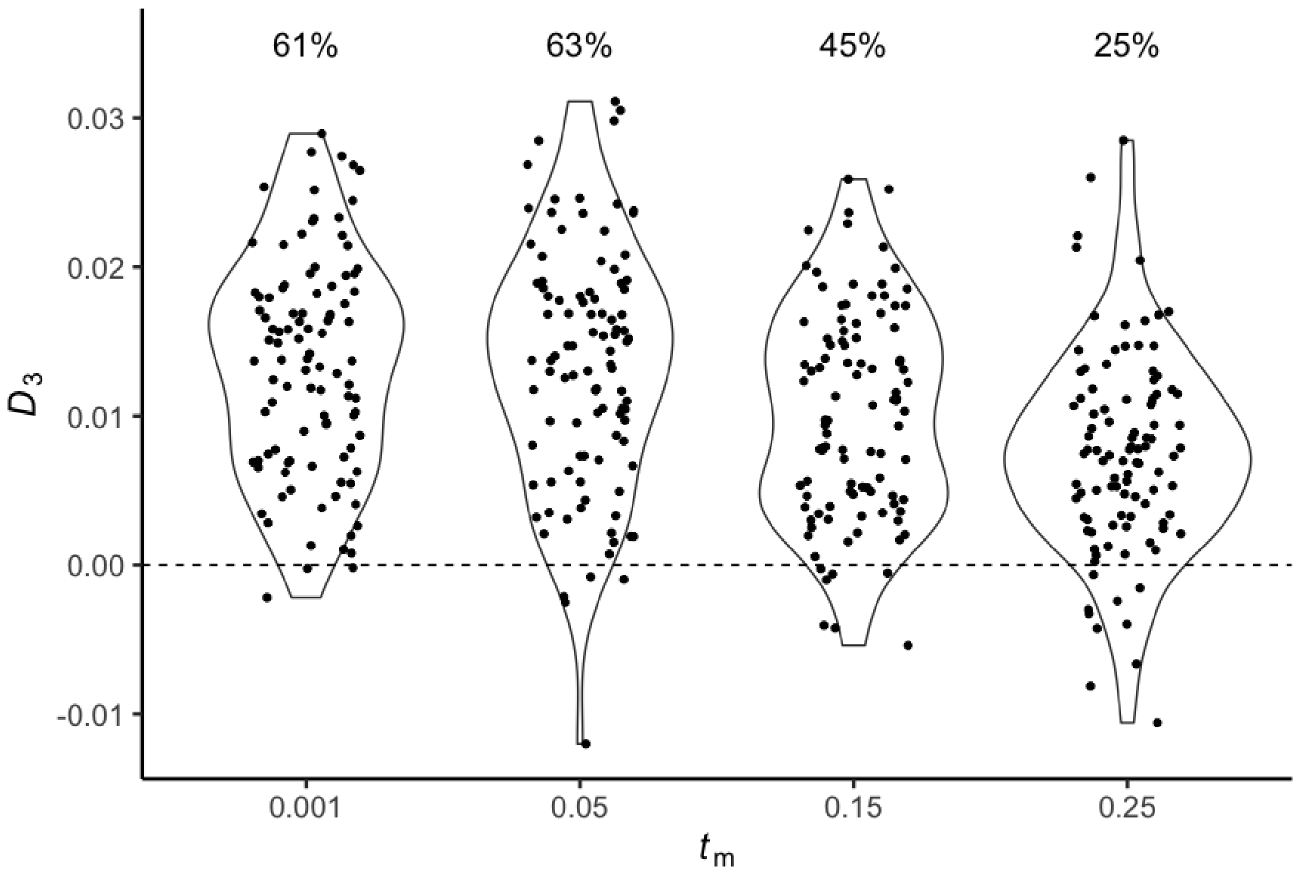

FIG. 5. Power of D_3 across four values of t_m (in units of $4N$ generations), with $\gamma = 0.05$. Note that in these simulations $t_1 = 0.3$, so that $t_m = 0.25$ is very shortly after speciation.

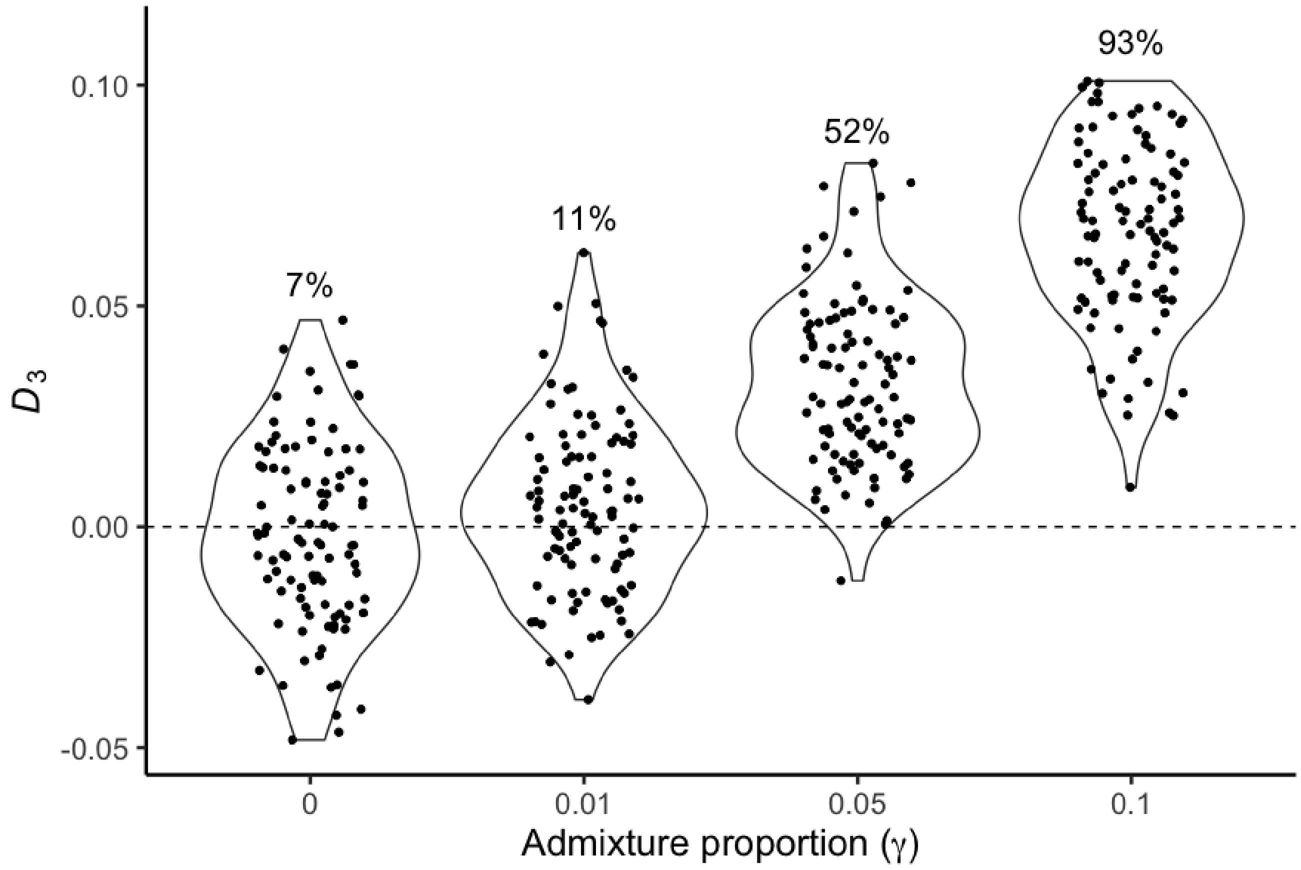

FIG. 6. Power of an alternate version of D_3 where d_{A-B} is used as the denominator instead of $(d_{B-C}+d_{A-C})$.

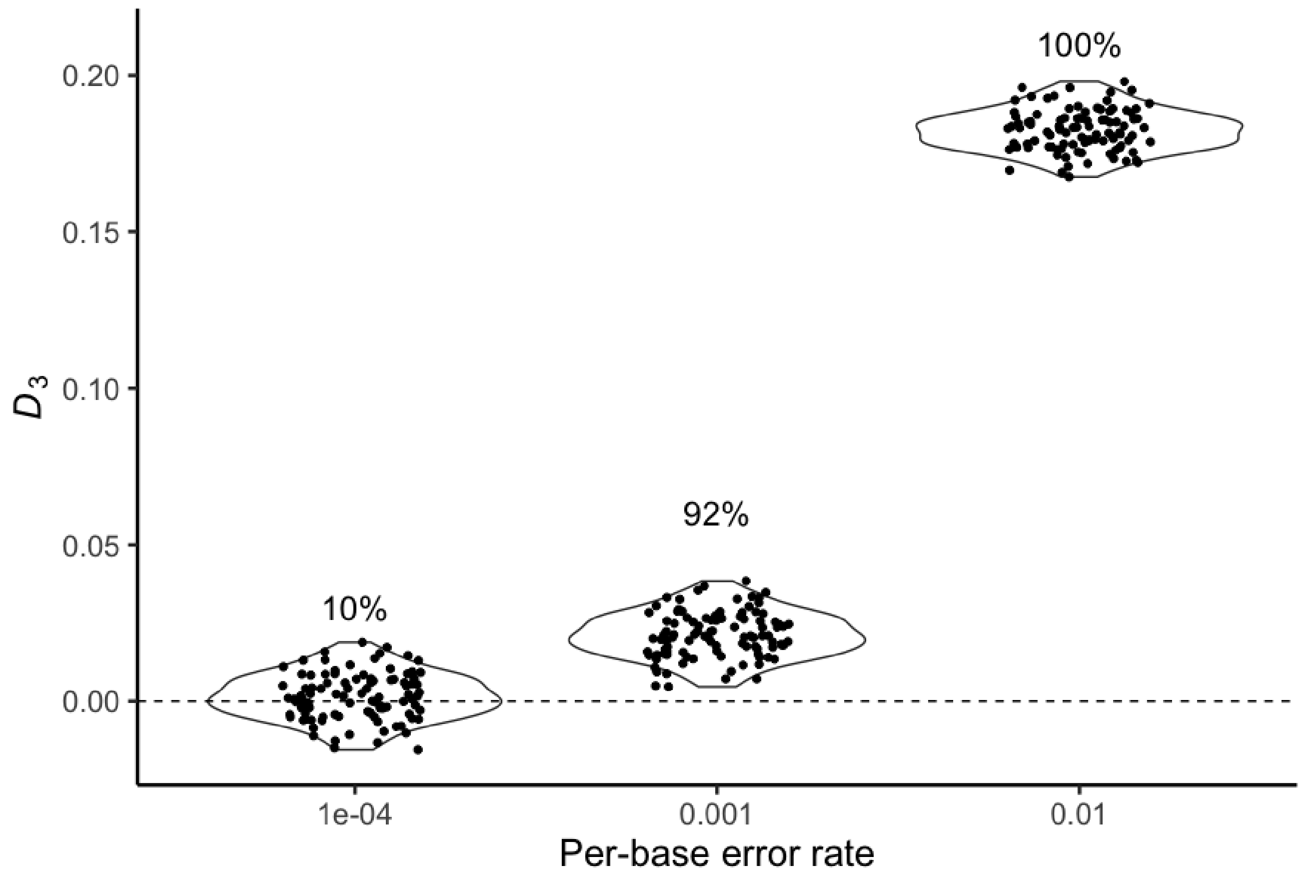

FIG. 7. False positive rate of D_3 ($\gamma = 0$) across different per-base error rates, with errors introduced to taxon B.

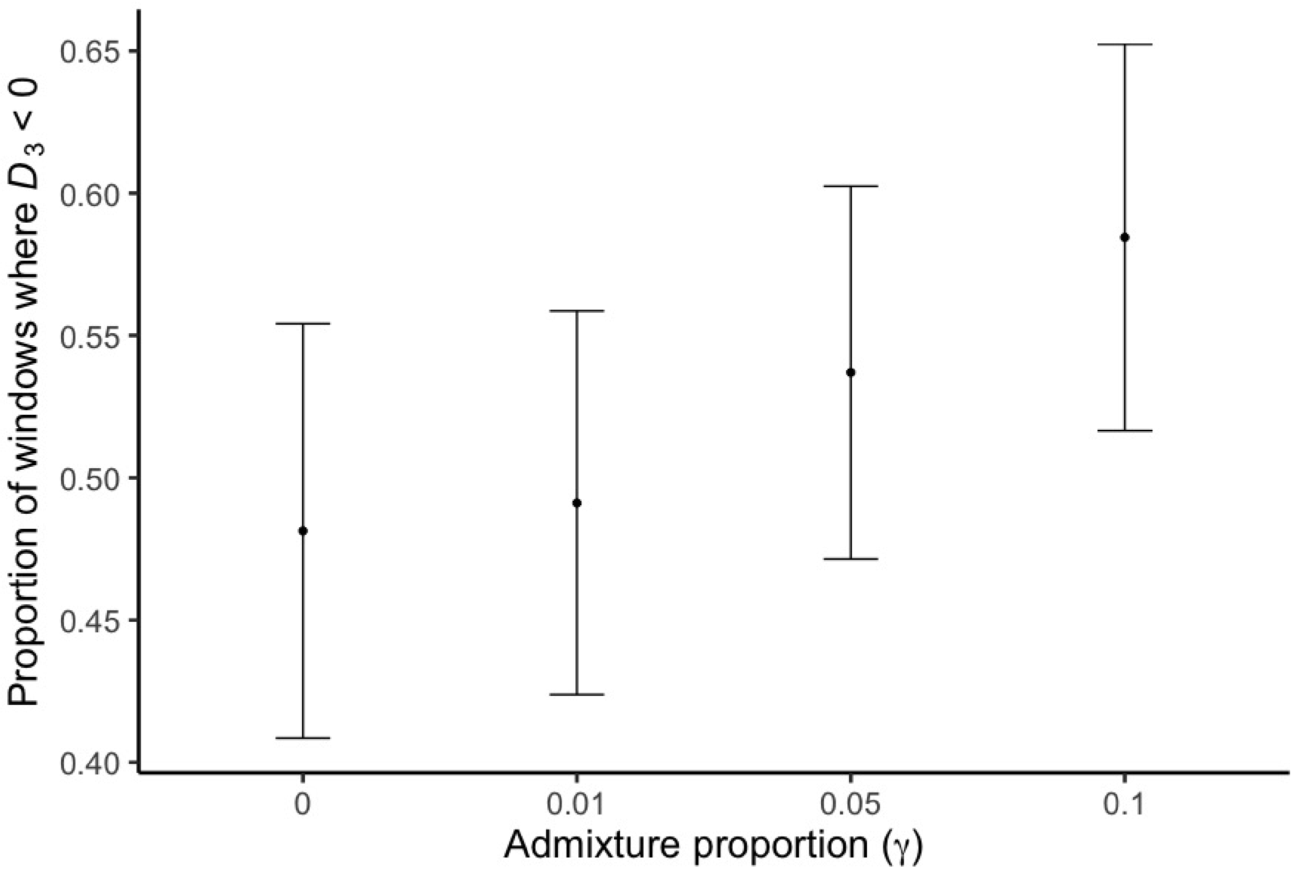

FIG. 8. Proportion of 5 kilobase alignment windows supporting the signal of introgression ($D_3 < 0$), as a function of the admixture proportion. Each point shows the mean and standard deviation across 100 simulated datasets. The expected value with no admixture is 0.50.